

FAGARAS, Romania

WMO: 152350

Lat: 45.8364N

Lon: 24.9353E

Elev: 429

StdP: 96.27

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-14.2	-19.1	0.7	-17.6	-15.6	1.0	-13.6	10.1	-0.6	8.4	1.6	0.3	70	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	30.6	20.5	29.0	19.7	27.5	19.2	22.6	27.4	21.4	26.6	20.5	25.5	1.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.3	16.8	24.3	19.9	15.4	23.0	18.9	14.5	22.3	68.9	27.1	64.4	26.8	60.9	25.3	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	7.3	5.8	-22.2	33.0	4.9	1.8	-25.7	34.4	-28.6	35.4	-31.4	36.5	-34.9	37.8		
(5)			-23.3	24.7	4.9	1.1	-26.9	25.5	-29.7	26.1	-32.5	26.7	-36.0	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-2.9	0.2	4.2	9.8	14.6	18.5	19.5	19.5	15.0	9.5	3.9	-1.7	(6)
(7)		DBStd	8.91	5.11	4.83	4.02	4.22	3.16	2.83	2.48	2.70	3.83	3.63	4.38	5.42	(7)
(8)		HDD10.0	1522	400	276	181	54	4	0	0	0	4	54	186	363	(8)
(9)		HDD18.3	3487	658	509	437	256	121	32	17	19	111	274	432	622	(9)
(10)		CDD10.0	1238	0	0	2	48	146	254	296	294	154	38	4	0	(10)
(11)		CDD18.3	160	0	0	0	0	3	36	54	56	11	0	0	0	(11)
(12)		CDH23.3	1941	0	0	0	24	99	394	558	655	202	9	0	0	(12)
(13)		CDH26.7	509	0	0	0	1	12	89	148	209	49	0	0	0	(13)
(14)	Wind	WSAvg	1.8	1.5	1.8	2.3	2.4	1.9	1.7	1.6	1.5	1.8	1.7	1.5	1.6	(14)
(15)	Precipitation	PrecAvg	707	32	31	39	56	87	103	101	73	61	52	38	37	(15)
(16)		PrecMax	974	71	58	69	112	181	207	189	158	148	99	92	100	(16)
(17)		PrecMin	443	8	4	7	17	34	27	25	15	10	3	1	4	(17)
(18)		PrecStd	128	15	14	16	23	40	46	41	43	30	26	21	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	15.4	20.0	25.7	28.2	31.1	32.0	32.9	30.5	24.2	19.3	10.1	(19)
(20)			MCWB	6.4	9.9	11.2	14.9	16.9	21.5	21.1	21.6	19.7	15.4	12.5	6.9	(20)
(21)		2%	DB	6.7	11.7	16.6	23.4	25.8	29.0	30.1	30.7	28.0	22.2	15.8	7.7	(21)
(22)			MCWB	5.0	7.6	9.6	13.8	16.7	20.1	20.8	20.2	18.1	14.3	11.4	6.0	(22)
(23)		5%	DB	4.3	9.0	13.8	20.4	24.1	27.3	28.3	28.9	25.6	19.7	13.2	5.7	(23)
(24)			MCWB	3.1	6.3	8.2	13.0	16.2	19.3	20.5	19.4	17.3	13.4	10.1	4.6	(24)
(25)		10%	DB	2.7	6.8	11.1	18.0	22.2	25.6	26.7	27.1	23.2	17.0	10.8	4.1	(25)
(26)			MCWB	1.9	4.9	7.1	11.9	15.5	18.8	20.0	19.1	16.6	12.5	8.4	3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	10.4	12.1	16.4	19.7	23.6	24.9	23.9	21.3	17.4	13.6	7.5	(27)
(28)			MCDWB	8.6	14.6	19.6	21.9	25.1	28.6	26.3	28.4	26.7	21.4	17.3	9.0	(28)
(29)		2%	WB	5.1	8.2	10.3	14.9	18.2	21.6	22.7	22.2	19.7	15.5	11.9	6.2	(29)
(30)			MCDWB	6.6	11.1	15.5	21.5	23.2	26.3	26.9	28.2	25.6	20.5	14.9	7.8	(30)
(31)		5%	WB	3.3	6.7	9.0	13.7	17.1	20.4	21.6	20.8	18.4	14.3	10.4	4.8	(31)
(32)			MCDWB	4.1	8.6	13.2	19.4	21.9	25.1	26.5	26.5	23.3	18.7	12.8	5.8	(32)
(33)		10%	WB	2.1	5.2	7.7	12.6	16.1	19.5	20.7	19.9	17.4	13.1	8.7	3.6	(33)
(34)			MCDWB	2.7	6.7	10.6	17.1	20.7	24.2	25.5	25.3	22.0	16.3	10.5	4.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	9.6	10.3	12.2	11.9	12.0	12.5	13.6	12.7	12.7	9.6	7.2	(35)
(36)			MCDBR	8.8	12.8	16.4	17.6	16.3	15.1	16.0	17.1	17.6	18.7	14.4	8.7	(36)
(37)			MCWBR	6.9	8.7	9.6	8.6	7.8	7.7	8.1	8.0	8.9	10.3	10.3	6.4	(37)
(38)		5% WB	MCDBR	8.4	12.2	14.6	15.2	14.0	13.8	15.0	15.8	16.1	15.2	13.4	7.6	(38)
(39)			MCWBR	6.8	8.6	9.0	8.1	7.6	7.7	8.4	8.1	8.9	9.3	9.7	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.333	0.375	0.432	0.415	0.413	0.431	0.441	0.397	0.359	0.334	0.306	(40)	
(41)		taud	2.332	2.275	2.234	2.118	2.234	2.275	2.232	2.200	2.294	2.391	2.406	2.371	(41)	
(42)		Ebn at Noon	789	833	845	824	854	855	833	807	814	796	751	747	(42)	
(43)		Edh at Noon	77	99	120	147	135	130	135	134	112	88	72	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.36	2.20	3.27	4.30	5.26	5.69	5.74	5.22	3.79	2.62	1.52	1.09	(44)	
(45)		RadStd	0.15	0.26	0.35	0.46	0.51	0.51	0.49	0.45	0.39	0.33	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (78 neighbors)		+0.60	N/A	N/A	+0.72	N/A	N/A	-128	-174	+76	+24

Nomenclature: See separate page